5.1 nth Roots and Rational Exponents

Name _____ Period ____ Date _____

Algebra II 5.1 · Student sheet

I can…

- find nth roots of numbers.
- evaluate expressions with rational exponents.
- solve equations using nth roots.

Vocabulary · fill in the missing word

nth root of a — a number b with b^n = _____

Index — _____

Rational exponent — $a^{\frac{1}{n}}$ = _____ and $a^{\frac{m}{n}} = (\sqrt[n]{a})^m = \sqrt[n]{a^m}$

Part 1 · How many real roots?

Find every real number that makes each equation true.

Equation	Real solution(s)	How many?
$x^2 = 16$		
$x^3 = -8$		
$x^4 = 81$		
$x^4 = -16$		
$x^5 = 32$		
$x^5 = -32$		

Rule: an equation $x^n = a$ has two real solutions when _____, one when _____, and none when _____

By the Power of a Power Property, $(9^{1/2})^2 = 9^1 = 9$. So $9^{1/2}$ must be _____. In the same way, $8^{1/3}$ must be _____.

Part 2 · Notes: fill in the blanks

If n is **odd**, a has exactly _____ real nth root.

If n is **even** and $a > 0$, a has _____ real nth roots. If n is even and $a < 0$, a has _____ real nth roots.

$a^{1/n}$ = _____ $a^{m/n}$ = _____ = _____

$a^{-m/n}$ = _____

In $a^{m/n}$, the denominator n is the _____ and the numerator m is the _____.

Part 3 · Examples (work them with the class)

Example 1

Find the indicated real nth root(s) of a.

a. $n = 3, a = -125$ b. $n = 4, a = 16$

Example 2

Evaluate each expression.

a. $16^{\left(\frac{1}{4}\right)}$ b. $(-27)^{\left(\frac{1}{3}\right)}$ c. $9^{\left(\frac{3}{2}\right)}$ d. $32^{\left(-\frac{3}{5}\right)}$

Example 3

a. Write $7^{\left(\frac{3}{4}\right)}$ in radical form. b. Write $\sqrt[5]{x^2}$ with a rational exponent.

Example 4

Find the real solution(s) of each equation. Round to two decimal places, if necessary.

a. $4x^5 = 128$ b. $(x - 3)^4 = 21$

Example 5

The radius r of a sphere with volume V is $r = \left(\frac{3V}{4\pi}\right)^{\left(\frac{1}{3}\right)}$. A basketball has a volume of about 455 cubic inches. Find its radius to the nearest hundredth of an inch.

Part 4 · Partner practice

Partner 1

Evaluate $27^{\left(\frac{2}{3}\right)}$ and $64^{\left(-\frac{1}{2}\right)}$.

Partner 2

Find the real solution(s) of $x^4 - 20 = 61$.